

Concept of Internet

The Internet is a global communication system that connects thousands of individual computer networks and forms a vast network of networks.

- It allows computers and devices around the world to exchange data and communicate.
- It supports services such as the Web, email, file transfer, online meetings and social media.

Internet = the worldwide network infrastructure.

World Wide Web = a service that runs on the Internet.

Concept of World Wide Web (WWW)

The World Wide Web (WWW) is a system of interlinked web pages and resources that users access through the Internet using web browsers.

Development and History of WWW

- The World Wide Web was invented by British computer scientist Tim Berners-Lee at CERN in 1989.
- The original idea was to make information sharing easier among researchers and institutions.
- By the end of 1990, the first Web server and browser were running at CERN.
- The first website was hosted at info.cern.ch.
- In 1991, Berners-Lee released the WWW software more widely, helping the Web spread beyond CERN.
- In 1993, CERN placed the Web software in the public domain, which helped accelerate its growth.

Uses of WWW

- Finding information and educational resources
- Reading news and digital documents
- Online shopping and banking
- Social networking and communication
- Streaming videos and multimedia
- Using web applications and online services

Web Browser

A web browser is a software application used to access, retrieve and display web pages and other web resources on the Internet.

Why do we need a Web Browser?

- Open websites using URLs.
- Display text, images, audio and video.
- Run interactive web applications.
- Navigate between linked pages.
- Provide features such as tabs, history, bookmarks and downloads.

Examples of Web Browsers

Browser	Developer/Organization	Common use
Google Chrome	Google	General web browsing
Mozilla Firefox	Mozilla	General browsing and privacy-focused use
Microsoft Edge	Microsoft	Windows and general browsing
Safari	Apple	Apple devices
Opera	Opera	General browsing

Search Engine

A search engine is an online system that searches its index of web information and returns results related to a user query.

Work of a Search Engine

- Crawls or discovers web pages and other resources.
- Indexes information so it can be searched quickly.
- Processes the user query and finds relevant matches.
- Ranks results and displays them on a results page.

Examples of Search Engines

Search Engine	General description
Google Search	Widely used general search engine
Microsoft Bing	General search engine from Microsoft
Yahoo Search	Web search service
DuckDuckGo	Search engine focused on privacy
Baidu	Major search engine used in China
Yandex	Major search engine used especially in Russia and nearby markets

Advanced Search Techniques

Search engines provide operators and special symbols that can make searches more precise. Common techniques include:

Technique	How it is used	Example	Purpose
AND	Use AND between terms.	computer AND security	Find results containing both terms.
OR	Use OR between alternatives.	college OR university	Find either alternative.
NOT	Use NOT to exclude a term.	jaguar NOT car	Reduce unwanted results.
* (asterisk)	Use * as a wildcard in some search systems.	"best * in Nepal"	Represents an unknown or variable word/part.
site:	Restrict results to one website/domain.	site:gov.np education	Search mainly within the specified site/domain.
filetype:	Limit results to a file type.	filetype:pdf networking	Find files of the specified type.
Quotation marks	Put an exact phrase inside quotation marks.	"computer network"	Search for the exact phrase.
Unquoted search	Enter normal words without quotation marks.	computer network	Search for pages containing related terms; exact wording is not required.

Search operators can vary slightly between search engines. Use them to narrow results and reduce irrelevant pages.

URL (Uniform Resource Locator)

A URL is the address used to identify and locate a resource on the Internet, such as a web page, image or file.

Key Components of a URL

Component	Meaning
Scheme / Protocol	Specifies how the resource should be accessed, such as http or https.
Domain	Identifies the website or host, such as example.com.
Top-Level Domain (TLD)	The ending of a domain, such as .com, .org or .np.
Path	Identifies a specific page or resource on the website.

<https://www.example.com/about/team>

Part	Example	Function
Scheme	https://	Uses secure HTTP communication.
Subdomain	www	Optional subdomain.
Domain	example	Main domain name.
TLD	.com	Top-level domain.
Path	/about/team	Points to a specific resource/page.

Concept of Remote Login / Remote Access

Remote access is the ability to connect to and operate another computer or device from a different location through a network or the Internet.

Examples: AnyDesk and TeamViewer

Feature	AnyDesk	TeamViewer
Main purpose	Remote desktop access, support and administration.	Remote support, meetings, file transfer and administration.
Best suited for	Quick, lightweight remote control and technical support.	Support plus broader collaboration and managed remote access.
Typical situation	A technician remotely fixes a computer for a user.	A support team remotely assists users or manages devices and collaborates.
Basic requirement	Both sides normally install/run the client and exchange the required access details.	Both sides normally install/run the client and exchange the required access details.

When to use them

- Use AnyDesk when you need simple, quick and generally lightweight remote desktop support.
- Use TeamViewer when remote support is combined with broader collaboration, device management or organization-level remote access.

Never give remote-control access to an unknown or untrusted person. Remote access can expose files, settings and personal information.

Concept of Web Protocols

A web protocol is a set of rules that defines how clients and servers communicate and exchange information on the Web.

HTTP (HyperText Transfer Protocol)

HTTP is an application-layer protocol used for communication between a client, such as a web browser, and a web server.

- A browser sends an HTTP request to a server.
- The server processes the request and sends an HTTP response.
- HTTP is used to transfer web resources such as HTML documents, images, scripts and other data.
- Ordinary HTTP traffic is not encrypted.

HTTPS (HyperText Transfer Protocol Secure)

HTTPS is HTTP carried over a secure TLS connection so that data exchanged between client and server is encrypted.

- Protects data from being easily read while in transit.
- Helps protect passwords, payment details and other sensitive information.
- Used widely for modern websites and web applications.

Difference between HTTP and HTTPS

Basis	HTTP	HTTPS
Meaning	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Security	Not encrypted by default	Encrypted using TLS
URL scheme	http://	https://
Typical use	Non-sensitive or legacy communication	Modern websites, login pages and sensitive transactions
Data protection	Lower	Higher

Email and Its Uses

Email (electronic mail) is a method of sending and receiving digital messages and files over a computer network or the Internet. Ray Tomlinson is widely credited with developing the first networked email system and introducing the @ symbol to separate the user name from the destination host.

Why do we use Email?

- Fast communication across distances.
- Sending documents and attachments.
- Formal communication for schools and workplaces.
- Record keeping and message history.
- Account registration and online notifications.

Uses of Email

Area	Uses
Personal communication	Messages, invitations, sharing photos/documents
Education	Assignments, notices, learning materials, communication with teachers
Business	Official communication, quotations, documents and coordination
Marketing	Newsletters, offers and customer communication
Entertainment	Subscriptions, event notifications and media updates

Social Media

Social media refers to online platforms that allow users to create, share, discuss and interact with content and with other users.

Advantages / Pros of Social Media

Advantage	Explanation
Global connection	Connect with people across countries and communities.
Information sharing	Share news, knowledge, educational material and ideas.
Business opportunities	Promote products, services and personal brands.
Diverse content	Access text, images, audio, video and live content.
Self-expression	Share opinions, creativity, skills and interests.
Real-time updates	Receive fast updates about events, communities and services.

Disadvantages / Cons of Social Media

Problem	Explanation
Isolation and loneliness	Excessive online interaction may reduce healthy offline interaction.

Misinformation	False or misleading information can spread quickly.
Cyberbullying	Users may face harassment, threats or abusive behavior.
Mental health impact	Heavy or unhealthy use may increase stress, comparison or anxiety.
Productivity loss	Excessive scrolling can waste time and reduce concentration.
Body image issues	Edited images and unrealistic standards may create unhealthy comparisons.

Proper Use of Social Media

- Use strong, unique passwords and enable multi-factor authentication where available.
- Do not share unnecessary personal information such as passwords, OTPs or private documents.
- Verify information before liking, sharing or forwarding it.
- Communicate respectfully and do not participate in cyberbullying.
- Review privacy settings and control who can see your posts and profile.
- Think before posting because online content can be copied or shared.
- Limit screen time and maintain a balance between online and offline activities.
- Report and block harmful, abusive or suspicious accounts when necessary.

Concept of Online Video Meeting

An online video meeting is a real-time communication session in which participants use the Internet to communicate using video, audio, chat and screen sharing.

Short History of Video Meetings

- Early video-communication experiments were conducted by Bell Labs in the early 20th century.
- AT&T demonstrated the Picturephone publicly at the 1964 New York World's Fair, including a transcontinental video call demonstration between New York and Disneyland.
- Modern Internet-based video meetings became practical with faster networks, webcams and cloud software.
- Services such as Zoom, Google Meet and Microsoft Teams later made online meetings common for education, business and remote work.

Zoom

- Zoom is a cloud-based communication platform used for video meetings, webinars, chat and collaboration.
- Zoom was founded by Eric Yuan in 2011, and Zoom Meetings had its first public release in 2013.

Google Meet

- Google Meet is Google's video meeting service for communication and collaboration.
- It grew from Google Hangouts Meet, introduced for enterprise video meetings in 2017.

Microsoft Teams

- Microsoft Teams is a collaboration platform that combines chat, meetings, file sharing and Microsoft 365 integration.
- Teams was announced in 2016 and became generally available in March 2017.

Platform	Main strengths	Common uses
Zoom	Video meetings, webinars, screen sharing	Classes, training, webinars, business meetings
Google Meet	Browser-based meetings, Google Workspace integration	Classes, school meetings, organizational communication
Microsoft Teams	Chat, meetings, channels, file collaboration, Microsoft 365 integration	Business teamwork, school collaboration, project work

Concept of Blog

A blog is a website or part of a website where an author or organization regularly publishes articles, opinions, news, guides, experiences or other content, usually with newer posts shown first.

Why do we need a Blog?

- Share knowledge, ideas and experiences.
- Publish educational or informational articles.
- Promote products, services or a personal brand.
- Create a record of opinions, projects or activities.
- Build a community through comments and subscriptions.

How does a Blog work?

1. The author creates or logs in to a blogging platform or content management system.
2. A post is written and may include images, videos or links.
3. The post is published on the blog and stored on a server/database.
4. Readers open the blog through a web browser and read the post.
5. Readers may comment, share or subscribe when those features are enabled.

Short History of Blogging

- Early personal web pages in the 1990s were similar to online journals and frequently updated personal sites.
- Jorn Barger is widely credited with popularizing the term "weblog" in 1997.
- Peter Merholz shortened "weblog" to "blog" in 1999.
- Blogging tools such as Blogger made online publishing easier for people who did not want to code a website from scratch.

Features of a Blog

Feature	Explanation
Posting	Allows authors to publish new articles or entries.
Authorship	Identifies the writer or organization that created the post.
Comments	Allows readers to respond and discuss when enabled.
Categories	Groups related posts under broad topics.
Archive	Organizes older posts by date or period.
Tags	Adds keywords that describe a post and help users find related posts.
Images and multimedia	Supports photos, videos, audio, embeds and other media.
Social sharing	Allows posts to be shared through social platforms or links.
Subscribe	Lets readers receive updates about new posts.